AMA572 Fixed-Income and Credit Risk

Revised Midterm Test — Worked Edition

Five questions, 20 marks each | Suggested time: 120 minutes

Instructions. Attempt all questions. Parts requiring proofs retain the mathematical statements of the supplied midterm. Numeric inputs in Questions 1 and 2 have been revised. Computations retain unrounded intermediate precision.
Important: The supplied solution to Question 5(b) requires an additional hypothesis. A counterexample and the precise mathematical limitation are provided instead of an invalid proof.

Contents

Question 1 (20 marks): Rating transitions **2**

Question 2 (20 marks): Implied default parameters **4**

Question 3 (20 marks): Tax shield and bankruptcy costs **6**

Question 4 (20 marks): Leland–Toft firm and debt values **7**

Question 5 (20 marks): Smooth pasting and the default barrier **9**

Question 1 (20 marks): Rating transitions

Given the following information:

(a) There are four credit classes A, B, C, D, where D is default (absorbing). Initially, there are **24** firms in each of A, B, C.

(b) After one month, **two** B firms migrate to C and **two** B firms migrate to A; they remain there during the rest of the year.

(c) After three months, **two** A firms migrate to B, and remain there.

(d) After seven months, **three** C firms migrate to B, and **two** C firms default.

(e) After ten months (measured from the start of the year), **two** further C firms default.

Compute (1) the empirical one-year transition matrix, (2) the estimated generator $\widehat{\Lambda}$, and (3) the generator-based one-year transition matrix.

Detailed solution

Step 1: terminal transitions by initial cohort. Initially-A: 22 stay A, two move to B. Initially-B: two move to A, 20 stay B, two move to C. Initially-C: three move to B, 17 stay C, four default. Thus, in state order (A, B, C, D),

$$\boxed{\widehat{P}_{\text{empirical}} = \begin{pmatrix} 22/24 & 2/24 & 0 & 0 \\ 2/24 & 20/24 & 2/24 & 0 \\ 0 & 3/24 & 17/24 & 4/24 \\ 0 & 0 & 0 & 1 \end{pmatrix}}.$$

Step 2: total exposure time in years. Let N_i be total years-at-risk spent in rating i by all firms; individuals that migrate contribute exposure to multiple ratings. The time of each event is measured from the initial date:

$$N_A = 22(1) + 2\left(\frac{3}{12}\right) + 2\left(\frac{11}{12}\right) = \boxed{\frac{73}{3} = 24.333333},$$

$$N_B = 20(1) + 4\left(\frac{1}{12}\right) + 2\left(\frac{9}{12}\right) + 3\left(\frac{5}{12}\right) = \boxed{\frac{277}{12} = 23.083333},$$

$$N_C = 17(1) + 5\left(\frac{7}{12}\right) + 2\left(\frac{10}{12}\right) + 2\left(\frac{11}{12}\right) = \boxed{\frac{281}{12} = 23.416667}.$$

Here 4 in N_B represents the four original B-rated firms leaving after month 1; 5 in N_C represents the three migrations and two defaults at month 7. The two firms arriving in C at month 1 contribute 11/12 year each. **Step 3: generator estimation.** For $i \neq j$, use $\widehat{\lambda}_{ij} = M_{ij}/N_i$, and set $\widehat{\lambda}_{ii} = -\sum_{j \neq i} \widehat{\lambda}_{ij}$. The event counts are $M_{AB} = 2$, $M_{BA} = 2$, $M_{BC} = 2$, $M_{CB} = 3$, $M_{CD} = 4$. Hence

$$\boxed{\widehat{\Lambda} = \begin{pmatrix} -6/73 & 6/73 & 0 & 0 \\ 24/277 & -48/277 & 24/277 & 0 \\ 0 & 36/281 & -84/281 & 48/281 \\ 0 & 0 & 0 & 0 \end{pmatrix}}.$$

Step 4: matrix exponential. For a one-year horizon $\Delta t = 1$, the continuous-time Markov estimator is $e^{\widehat{\Lambda}\Delta t}$. Numerically,

$$\widehat{P}_{\text{generator}} \approx \begin{pmatrix} 0.924282 & 0.072574 & 0.002967 & 0.000177 \\ 0.076504 & 0.848474 & 0.068679 & 0.006344 \\ 0.004625 & 0.101552 & 0.745910 & 0.147913 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Checks. Rows sum to one (up to rounding). D is absorbing. Unlike the empirical matrix, the generator matrix produces small positive probabilities for indirect paths such as $A \to B \to C$.

Question 2 (20 marks): Implied default parameters

A company issues three- and five-year bonds with an annual coupon of **4.2%** of par, paid annually. Their continuously compounded yields are **4.6%** and **4.9%**, respectively. The flat continuously compounded risk-free rate is **2.7%**; recovery at default is **45%** of par. Assume default can occur only halfway through a year, at $0.5, 1.5, 2.5, \ldots$. Using the course's expected-loss approximation, calculate P_1 for years 1–3 and P_2 for years 4–5. Take par value as 100.

Detailed solution

Valuation convention. The course's midterm approximates default losses by multiplying the discounted loss in each year by P_1 or P_2 and summing, without compounding survival probabilities. Accordingly, P_1, P_2 below are *piecewise annual default-probability parameters under that approximation*, not exact continuously compounded hazard intensities. For a maturity-T bond, the risk-free value of its remaining promised cash flows as seen at t (immediately before default) is

$$F_T^{\mathrm{rf}}(t) = \sum_{k=\lfloor t \rfloor+1}^{T} 4.2e^{-0.027(k-t)} + 100e^{-0.027(T-t)}.$$

The loss upon default is $L_T(t) = F_T^{\mathrm{rf}}(t) - 45$, with time-zero present value factor $e^{-0.027t}$. **(I) Three-year bond.** Prices inferred from promised cash flows are

$$B_3^{\mathrm{rf}} = 4.2e^{-0.027} + 4.2e^{-2(0.027)} + 104.2e^{-3(0.027)} = \boxed{104.159915},$$
$$B_3^{\mathrm{risky}} = 4.2e^{-0.046} + 4.2e^{-2(0.046)} + 104.2e^{-3(0.046)} = \boxed{98.610502}.$$

t	$F_3^{\mathrm{rf}}(t)$	Recovery	Loss $L_3(t)$	e^{-rt}	Discounted loss
0.5	105.5756	45	60.5756	0.986591	59.7633
1.5	104.2079	45	59.2079	0.960309	56.8579
2.5	102.8028	45	57.8028	0.934728	54.0298

Therefore,

$$104.15991471 - 98.61050155 = (59.76333248 + 56.85788509 + 54.02983522)P_1,$$

$$\boxed{P_1 = \frac{5.54941316}{170.65105279} = 0.03251907 \approx 3.2519\%.}$$

(II) Five-year bond. Similarly,

$$B_5^{\mathrm{rf}} = \sum_{k=1}^{5} 4.2e^{-0.027k} + 100e^{-5(0.027)} = \boxed{106.751779},$$
$$B_5^{\mathrm{risky}} = \sum_{k=1}^{5} 4.2e^{-0.049k} + 100e^{-5(0.049)} = \boxed{96.443185}.$$

t	$F_5^{\mathrm{rf}}(t)$	$L_5(t)$	e^{-rt}	Discounted loss	Parameter
0.5	108.2027	63.2027	0.986591	62.3552	P_1
1.5	106.9069	61.9069	0.960309	59.4497	P_1
2.5	105.5756	60.5756	0.934728	56.6217	P_1
3.5	104.2079	59.2079	0.909828	53.8690	P_2
4.5	102.8028	57.8028	0.885591	51.1896	P_2

The first three rows sum to 178.42664706; the last two sum to 105.05858644. Since P_1 has already been estimated from the shorter bond,

$$\begin{aligned} 106.75177947 - 96.44318512 &= 178.42664706P_1 + 105.05858644P_2, \\ \boxed{P_2} &= \frac{10.30859435 - 178.42664706(0.03251906783)}{105.05858644} \\ &= \boxed{0.04289346 \approx 4.2893\%.} \end{aligned}$$

Economic check: Under the course approximation, the implied credit risk is greater in years 4–5 than in years 1–3: $P_2 > P_1$.

Question 3 (20 marks): Tax shield and bankruptcy costs

Consider the Black–Scholes–Merton geometric Brownian motion model

$$dV(t) = rV(t)\,dt + \sigma V(t)\,dW(t), \qquad V(0) = 1,$$

where r and σ are constants. Compute

$$A(0) = \mathbb{E}^*\left[e^{-rT}\kappa D \mathbf{1}_{\{V(T)\geq D\}}\right], \qquad C(0) = \mathbb{E}^*\left[e^{-rT}(1-\alpha)V(T)\mathbf{1}_{\{V(T)<D\}}\right].$$

Detailed solution

Step 1: solve the SDE. By Itô's lemma for $\log V(t)$,

$$d\log V(t) = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma\,dW(t).$$

Therefore (taking $V(0) = 1$),

$$V(T) = \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right\}, \quad Z \sim N(0,1),$$

and $X := \log V(T) \sim N(\mu, s^2)$, where $\mu = (r - \sigma^2/2)T$, $s = \sigma\sqrt{T}$. Let Φ denote the standard normal CDF. Then

$$\mathbb{P}^*(V(T) \geq D) = \mathbb{P}^*(X \geq \log D) = \Phi\left(\frac{\mu - \log D}{s}\right).$$

Step 2: tax benefit. Directly,

$$\boxed{A(0) = \kappa D e^{-rT}\Phi\left(\frac{(r-\sigma^2/2)T - \log D}{\sigma\sqrt{T}}\right).}$$

Step 3: bankruptcy cost. Write the truncated lognormal expectation as

$$\mathbb{E}^*[V(T)\mathbf{1}_{\{V(T)<D\}}] = \int_{-\infty}^{\log D} e^x \frac{1}{s\sqrt{2\pi}}\exp\left[-\frac{(x-\mu)^2}{2s^2}\right]dx.$$

Complete the square, using

$$x - \frac{(x-\mu)^2}{2s^2} = \mu + \frac{s^2}{2} - \frac{[x-(\mu+s^2)]^2}{2s^2}.$$

Consequently,

$$\begin{aligned}\mathbb{E}^*[V(T)\mathbf{1}_{\{V(T)<D\}}] &= e^{\mu+s^2/2}\Phi\left(\frac{\log D - \mu - s^2}{s}\right)\\ &= e^{rT}\Phi\left(\frac{\log D - (r+\sigma^2/2)T}{\sigma\sqrt{T}}\right).\end{aligned}$$

Multiplying by $e^{-rT}(1-\alpha)$ gives

$$\boxed{C(0) = (1-\alpha)\Phi\left(\frac{\log D - (r+\sigma^2/2)T}{\sigma\sqrt{T}}\right).}$$

Check. For $D \downarrow 0$, both quantities tend to zero. The tax shield is discounted by e^{-rT}, while the risk-neutral asset expectation cancels that factor in $C(0)$.

Question 4 (20 marks): Leland–Toft firm and debt values

The firm's asset value satisfies

$$\frac{dV_t}{V_t} = (r-\delta)dt + \sigma\, dW_t, \qquad V_t = V_0 \exp\Big\{(r-\delta-\sigma^2/2)t + \sigma W_t\Big\}.$$

(a) If $\tau = \inf\{t \geq 0 : V_t \leq V_B\}$ and

$$v(V;V_B) = V + \kappa\rho P\, \mathbb{E}\left[\int_0^\tau e^{-rt}dt\right] - (1-\alpha)V_B\mathbb{E}[e^{-r\tau}],$$

show that

$$v(V;V_B) = V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+2r/\sigma^2}+\xi}, \quad \xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2}.$$

(b) If $D(V;V_B) = P\int_0^\infty m e^{-mT} B(V;V_B,T)dT$ with recovery $c = (m+r)\alpha V_B/(mP)$, derive the explicit total debt value.

Detailed solution

Set $h_a := \sqrt{\xi^2 + 2a/\sigma^2} + \xi$. Because $\log(V_t/V) = (r-\delta-\sigma^2/2)t + \sigma W_t$ and the barrier is below V, Lecture 3's downward first-passage Laplace transform gives

$$\boxed{H(a) := \mathbb{E}[e^{-a\tau}\mathbf{1}_{\{\tau<\infty\}}] = \left(\frac{V_B}{V}\right)^{h_a}} \qquad (a>0).$$

(a) Firm value. Integrating pathwise and taking expectations,

$$\mathbb{E}\left[\int_0^\tau e^{-rt}dt\right] = \frac{1-\mathbb{E}[e^{-r\tau}]}{r} = \frac{1-H(r)}{r}.$$

Thus

$$\begin{aligned} v(V;V_B) &= V + \frac{\kappa\rho P}{r}(1-H(r)) - (1-\alpha)V_B H(r) \\ &= \boxed{V + \frac{\kappa\rho P}{r} - \left[\frac{\kappa\rho P}{r} + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^{h_r}}. \end{aligned}$$

(b) Debt value. With unit face value and continuously paid coupon ρ,

$$B(V;V_B,T) = \mathbb{E}\left[e^{-rT}\mathbf{1}_{\{\tau>T\}} + ce^{-rT}\mathbf{1}_{\{\tau\leq T\}} + \int_0^{\tau\wedge T}\rho e^{-rs}ds\right].$$

Interchanging nonnegative integrations in the aggregate bond valuation gives three components (write $h = H(m+r)$):

$$\begin{aligned} I_{\text{principal}} &= \int_0^\infty m e^{-(m+r)T}\mathbb{P}(\tau>T)dT = \frac{m}{m+r}(1-h), \\ I_{\text{coupon}} &= \rho\int_0^\infty m e^{-mT}\mathbb{E}\left[\int_0^{\tau\wedge T} e^{-rs}ds\right]dT = \frac{\rho}{m+r}(1-h), \\ I_{\text{recovery}} &= c\int_0^\infty m e^{-(m+r)T}\mathbb{P}(\tau\leq T)dT = \frac{mc}{m+r}h. \end{aligned}$$

For instance, the first equality uses $\int_0^\tau me^{-(m+r)T}dT = m(1-e^{-(m+r)\tau})/(m+r)$. For the coupon component, swapping the s and T integrals yields $\int_s^\infty me^{-mT}dT = e^{-ms}$. Combining all three yields

$$\begin{aligned} D(V;V_B) &= P\left[\frac{m+\rho}{m+r}(1-h) + \frac{mc}{m+r}h\right] \\ &= P\frac{m+\rho}{m+r} - \left[P\frac{m+\rho}{m+r} - \alpha V_B\right]h. \end{aligned}$$

Therefore,

$$\boxed{D(V;V_B) = P\frac{m+\rho}{m+r} - \left[P\frac{m+\rho}{m+r} - \alpha V_B\right]\left(\frac{V_B}{V}\right)^{h_{m+r}}.}$$

Expanding its coefficient gives exactly the more elaborate expression in the supplied examination:

$$D = P\frac{m+\rho}{m+r} - P\left[\frac{\rho}{r} + \left(1 - \frac{\rho}{r} - \frac{m+r}{m}\frac{\alpha V_B}{P}\right)\frac{m}{m+r}\right]\left(\frac{V_B}{V}\right)^{h_{m+r}}.$$

Question 5 (20 marks): Smooth pasting and the default barrier

Keep the original definition $S(V;V_B) = v(V;V_B) - D(V;V_B)$ from Question 4, with the same asset dynamics and notation. (a) Show that $\partial_V S(V;V_B^*)|_{V=V_B^*} = 0$ yields $V_B^* = \varepsilon P$, where

$$\varepsilon = \frac{\frac{\rho+m}{r+m}h_{r+m} - \frac{\kappa\rho}{r}h_r}{1+(1-\alpha)h_r + \alpha h_{r+m}}.$$

(b) Show that $0 < \varepsilon < (\rho+m)/(\alpha r+m)$.

Detailed solution

(a) Derive the barrier. The formulas from Question 4 show

$$\begin{aligned} S(V;V_B) = V + \frac{\kappa\rho P}{r} - \left[\frac{\kappa\rho P}{r} + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^{h_r} \\ - P\frac{m+\rho}{m+r} + \left[P\frac{m+\rho}{m+r} - \alpha V_B\right]\left(\frac{V_B}{V}\right)^{h_{m+r}}. \end{aligned}$$

Hold the candidate barrier V_B fixed when differentiating with respect to current assets V. For $k > 0$, $\partial_V (V_B/V)^k = -(k/V)(V_B/V)^k$. Hence

$$\begin{aligned} \frac{\partial S}{\partial V} = 1 + \frac{h_r}{V}\left[\frac{\kappa\rho P}{r} + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^{h_r} \\ - \frac{h_{m+r}}{V}\left[P\frac{m+\rho}{m+r} - \alpha V_B\right]\left(\frac{V_B}{V}\right)^{h_{m+r}}. \end{aligned}$$

At $V = V_B$, both powers equal one. Smooth pasting gives

$$0 = V_B + h_r\left[\frac{\kappa\rho P}{r} + (1-\alpha)V_B\right] - h_{m+r}\left[P\frac{m+\rho}{m+r} - \alpha V_B\right].$$

Collect the terms in V_B:

$$V_B[1 + (1-\alpha)h_r + \alpha h_{m+r}] = P\left[\frac{m+\rho}{m+r}h_{m+r} - \frac{\kappa\rho}{r}h_r\right].$$

Therefore,

$$\boxed{V_B^* = \varepsilon P, \quad \varepsilon = \frac{\frac{m+\rho}{m+r}h_{m+r} - \frac{\kappa\rho}{r}h_r}{1+(1-\alpha)h_r + \alpha h_{m+r}}.}$$

(b) Validity audit of the supplied inequality. The requested bound is *not valid for all positive model parameters stated in the problem*; therefore it cannot be rigorously proved without additional assumptions. The original solution attempts to replace the denominator by αh_{m+r} and then concludes a smaller bound with $\alpha r + m$; that last comparison does not follow. A concrete counterexample consistent with $r > \kappa\rho$, $0 < \alpha < 1$, and $m > 0$ is

$$r = 0.05, \quad \delta = 0.03, \quad \sigma = 0.20, \quad \rho = 0.06, \quad \kappa = 0.25, \quad \alpha = 0.50, \quad m = 1.$$

Then $\xi = 0$, $h_r = \sqrt{2.5} = 1.581139$, $h_{m+r} = \sqrt{52.5} = 7.245688$, and

$$\varepsilon \approx 1.263593, \qquad \frac{\rho+m}{\alpha r+m} = \frac{1.06}{1.025} \approx 1.034146.$$

Thus $\varepsilon > (\rho+m)/(\alpha r+m)$, contradicting the requested inequality. The counterexample also has $\varepsilon > 0$ and satisfies $\kappa\rho = 0.015 < r$. **What can be proved without the missing assumption?** If $0 \le \kappa \le 1$, $0 < \rho < r/\kappa$, $m > 0$, and $0 < \alpha \le 1$, then $h_{m+r} > h_r > 0$ and

$$\frac{m+\rho}{m+r} > \frac{\kappa\rho}{r} \quad \text{if, for example, } 0 < \rho \le r \text{ and } 0 \le \kappa < 1,$$

which ensures the numerator is positive under the displayed coefficient restriction. Since the denominator exceeds αh_{m+r},

$$\boxed{0 < \varepsilon < \frac{m+\rho}{\alpha(m+r)}}$$

holds whenever the numerator is positive. This weaker upper bound is correct; it is not the stronger upper bound stated in the original exam. In general, positivity is guaranteed exactly when

$$\frac{m+\rho}{m+r} h_{m+r} > \frac{\kappa\rho}{r} h_r.$$

Any claim of the stronger bound needs an additional parameter restriction that ensures it.

Reference conventions

The methods used are the lecture's cohort exposure generator estimator (Lecture 1 / Assignment 1), the original midterm's expected-loss approximation (Lecture 1 / Assignment 1), geometric Brownian motion and truncated lognormal moments (Lecture 2), downward first-passage Laplace transforms (Lecture 3), and the Leland–Toft leverage and smooth-pasting framework (Lecture 4 / Assignment 2).